

Solving Logarithmic Equations with Domain Checks

Focus: Rewriting a logarithm in exponential form, equating like logarithms, combining logarithms before solving, and approximating solutions.

The four-step pattern. Follow it on every problem; the domain step is not optional, because the algebra that clears logarithms can create solutions the original equation never had.

Worked pattern: Solve $\log_4(x + 1) = 3$.

1. Domain. Every logarithm needs a positive argument: $x + 1 > 0$, so $x > -1$.

2. Rewrite. $\log_b A = c$ means $A = b^c$, so $x + 1 = 4^3 = 64$.

3. Solve. $x = 63$.

4. Check. $63 > -1$, so the candidate is in the domain, and $\log_4(64) = 3 \checkmark$. Answer: $x = 63$.

For each problem below: **state the domain restriction first**, solve, and check every candidate against that domain. Give exact answers unless the problem asks for a decimal.

1. Solve: $\log_2 x = 5$

Answer: _____

2. Solve: $\log_3(x - 4) = 2$

Answer: _____

3. Solve: $\log_5(2x + 1) = \log_5(x + 9)$

(Two logarithms with the same base are equal exactly when their arguments are equal — but only where both arguments are positive.)

Answer: _____

4. No solving here — build the tool. Use the product rule to rewrite $\log_2(8x)$ in the form $c + \log_2 x$ for a constant c , and state the values of x for which the rewriting is valid.

Answer: _____

5. Solve: $\log_2 x + \log_2(x - 2) = 3$

(Combine the left side into a single logarithm first.)

Answer: _____

6. Clearing the logarithms in problem 5 gives the quadratic $x^2 - 2x - 8 = 0$, whose roots are $x = 4$ and $x = -2$. Evaluate the argument $x - 2$ at the candidate $x = -2$, then use that value to explain in one sentence why -2 cannot be a solution of the original equation. State the domain of that equation as well.

Answer: _____

7. Solve exactly (leave e in your answer): $\ln(x - 1) = 2$

Answer: _____

8. Solve: $\log_2(x^2 + 4x) = \log_2 12$

(Careful — do not assume the negative candidate is extraneous. Test it against the domain before you decide.)

Answer: _____

9. Solve for x and round to the nearest hundredth: $\ln(2x - 1) = 3$

Answer: _____

10. Solve: $\log(x + 3) - \log(x - 1) = 1$
(Base-10 logarithms. Use the quotient rule first.)

Answer: _____

11. Solve for x and round to the nearest hundredth: $\log_2(x + 5) = 4.7$

Answer: _____

12. Synthesis. Solve: $2 \log_5 x - \log_5(x + 6) = 0$

(Power rule, then quotient rule, then remember what an exponent of 0 produces. State the domain before you start and say which candidate it eliminates.)

Answer: _____